

(magnetic force) is attractive so distance (between turns) decreases

B1

7 (a) number of cycles per unit time

B1

(b) (i) period = $2\pi / 40\pi = 0.050 \text{ s} = 50 \text{ ms}$

A1

(ii) sinusoidal curve, starting at (0, 0) and initially increasing from there

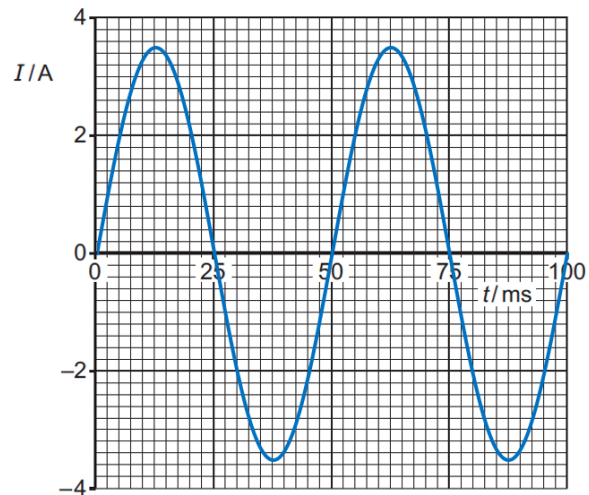
B1

periodic line showing 2 cycles with period 50 ms from $t = 0$ to $t = 100 \text{ ms}$

B1

all peaks shown at $I = +3.5 \text{ A}$ and all troughs shown at $I = -3.5 \text{ A}$

B1



$$\begin{aligned} \text{(iii)} \quad I_{\text{r.m.s}} &= 3.5/\sqrt{2} \\ &= 2.5 \text{ A} \end{aligned}$$

A1

$$\text{(c)} \quad P = I^2 R$$

$$\text{peak power} = 3.5^2 \times 680 (= 8330 \text{ W})$$

M1

or

$$\text{mean power} = 2.47^2 \times 680 (= 4170 \text{ W})$$

M1

peak and mean powers both calculated correctly, with supporting working, and compared leading to conclusion that mean power is half the peak power **A1**